EART27201— Sedimentary Rocks and Fossils
Fossil Practicals — Trilobites & Echinoderms

Russell Garwood | russell.garwood@manchester.ac.uk

Often there will be QR codes that link to online resources. If you are accessing the page via e.g. a laptop (so resolving the QR code is challenging), you can find a list of all the links at this URL:

https://www.russellgarwood.co.uk/EART27201.html

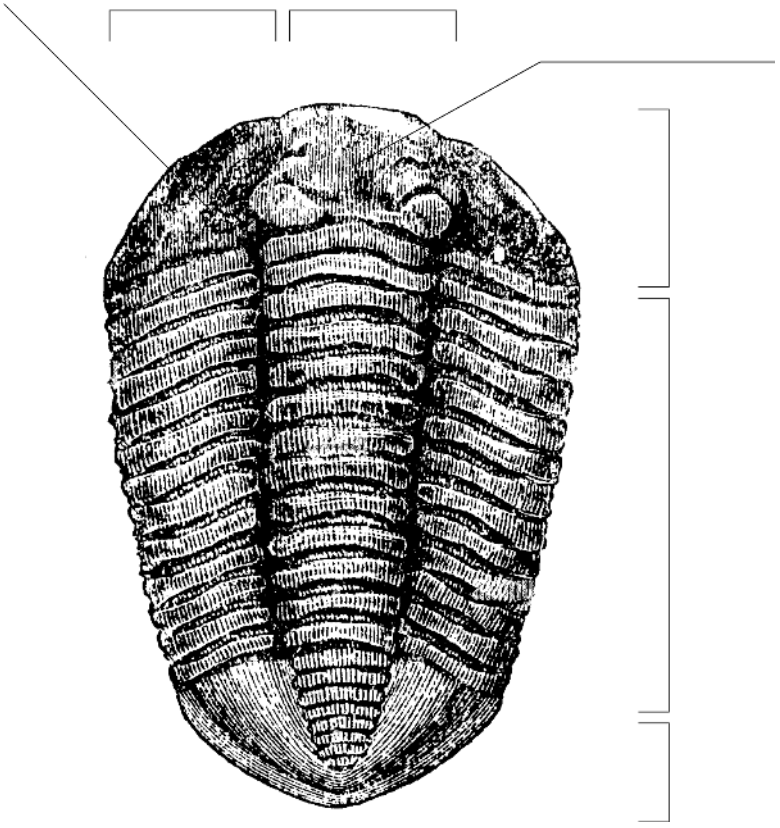
Under each QR code in this booklet is a keyword in italics - click the associated link on the page above, and it will take you to the correct target.

Greetings and salutations. Today we're looking at trilobites and then echinoderms. It'll be great.

Trilobites

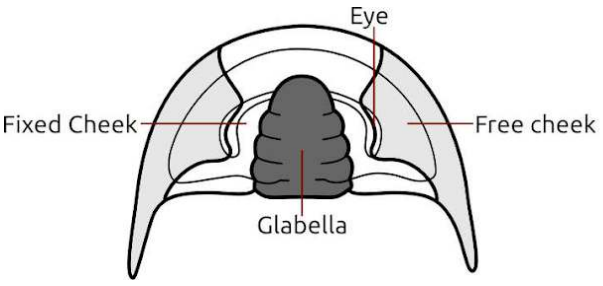
Let's start by meeting a trilobite in a bit more detail. You should have access to a specimen of a Silurian genus *Calymene* within arm's reach (a member of the order Phacopida, if you're interested). Below is a lovely 18th century woodcut of a member of this genus. Let's make sure we're happy with trilobite anatomy and go old school by labelling this.

Trilobites task 1: Study the hand specimen, and put the following labels in the correct place on our diagram: axial lobe, cephalon, eyes, glabella, pleural lobe, pygidium, thorax.



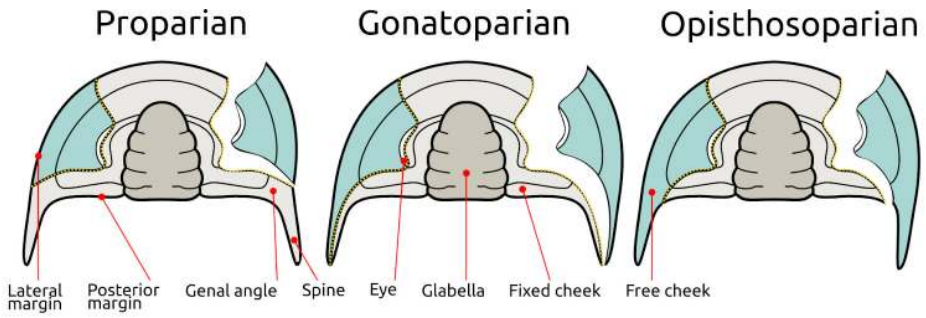
Let's think about facial sutures next. These were once really important for identifying the taxonomy of the group. This is a little less true today, as it's recognised they actually vary a lot, but it's worth meeting the three major flavours. Let's remind ourselves of the arrangement of a head, courtesy of this week's glossary:

Cephalon: The anterior or head region of a trilobite, which consists of at least five fused segments and is generally semicircular in shape. The glabella (the raised median part of the cephalon) is very variable in size and structure and it is probable that in life the stomach lay beneath it. Between the lateral margins of the glabella and the facial suture (see cephalic suture) is the fixed cheek (fixigena) and between the facial suture and the lateral margin is the free cheek (librigena). The glabella and the fixed cheeks together are known as the 'cranidium'. Above the visual surface of the eye is the palpebral lobe.



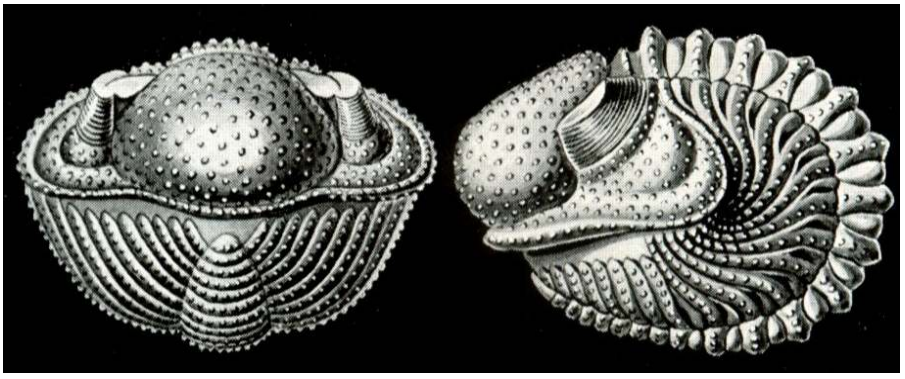
Facial (= cephalic) suture: In trilobites, the structure that includes the facial and ventral cephalic sutures. The facial sutures situated on the dorsal part of the cephalon are of three main types: protoparian, where the suture runs from the anterior border of the cephalon to a position anterior to the genal angle; opisthoparian, where the suture runs

from the anterior margin to cut the posterior margin posterior to the genal angle; and marginal, where the suture runs along the edge and is not visible on the upper surface. In one family the facial suture runs directly through the genal angle and is 'gonatoparian'.



Trilobites task 2: Check out the Siluian beast *Dalminites* in front of you (also a phacopid). The casts of Calymene aren't the best, so you can only see the facial sutures on a few of those. These ones are a little better, and the facial suture is particularly visible on the right hand side. Which of the above arrangements is this trilobite?

Thanks to their articulation, trilobites could enroll:



In early trilobites - i.e. Cambrian species - this ability was a bit limited, but as the groups evolved into the Ordovician it became much more common.

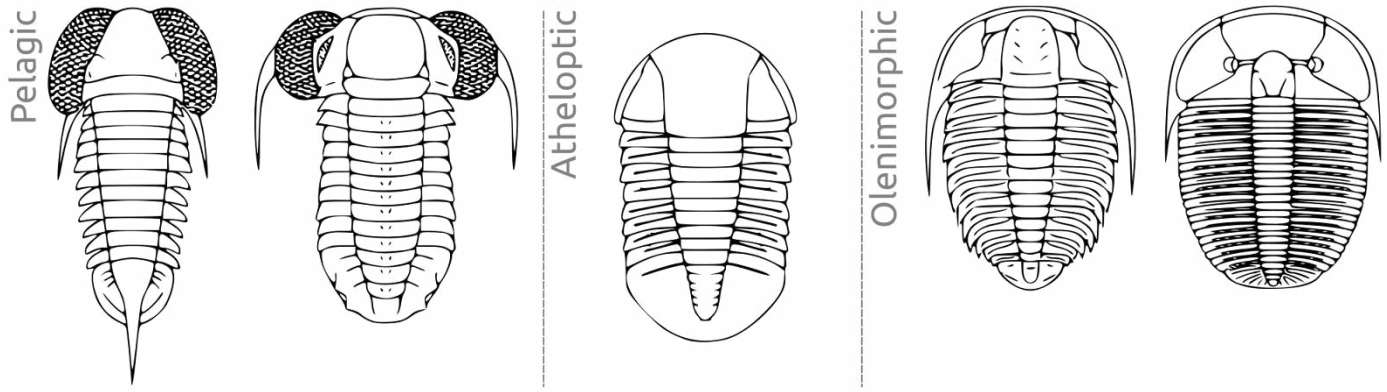
For a 3D model of an enrolled trilobite, check out this link:

Link *Flexicalymene*

Indeed, in the Ordovician, several groups evolved a variety of sophisticated structures to enhance this behaviour. An example is found in the order Phacopida which you met above - members of this order evolved to have tooth and socket pairs on their front and back end. These locked together to hold the trilobite in a tight ball.

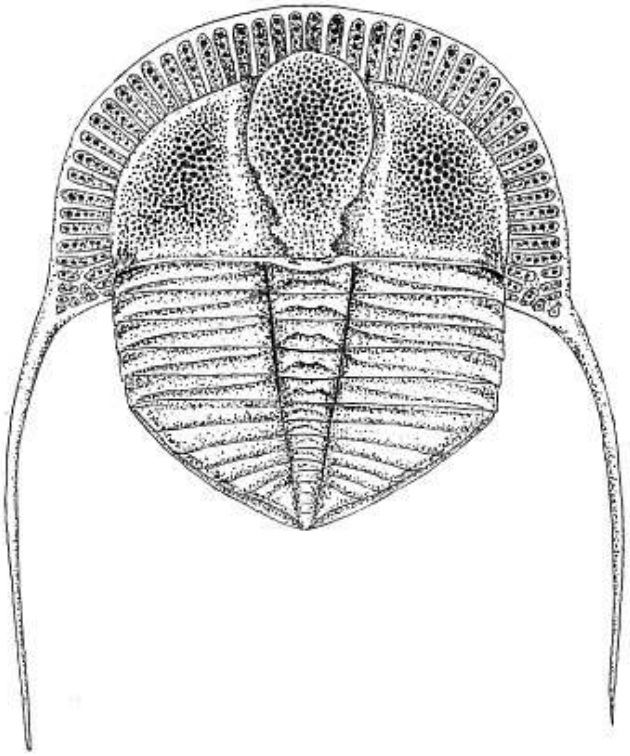
Trilobites task 3: Why do you think so many different trilobite groups evolved this ability? What advantage may enrollment have provided?

This is an example of convergent evolution - something we met in last week's practical in terms of the similarity between bivalves and brachiopods. It's not the only convergent evolution in trilobites, though. The group was very diverse in the Palaeozoic, as we have learned, and during this era there are many instances of convergent evolution based on the ecology (mode of life) of the trilobites. We can use their morphology, bolstered by evidence from the other organisms they are found with, the sediment in which they are preserved, and their trackways, to reconstruct this ecology. Morphologies associated with a particular mode of life are referred to as ecomorphs - some examples are shown below. Do bear in mind that these have evolved multiple times in different groups of trilobites.



For example, we may identify swimmers (pelagic organisms) that are suited to life in water column, possibly surface waters, on the basis of large (=hypertrophied) eyes and reduced pleurae. Atheloptic species had reduced eyes and thus may have inhabited deeper-water habitats. Olenimorphs with their thin exoskeletons may have been associated with life in dysaerobic environments. The table below (courtesy of Casey and Lieberman 2014) provides a brief summary of morphological traits, the interpretation of these, and the inferred mode of life built on those interpretations.

Fossil Evidence	**Interpretation**	**Inferred lifestyle or behaviour**
Reduced thorax and pygidium; smoothed cephalon; downward projecting spines; facies independence	Light, streamlined body allows fast swimming. Spines prohibit effective movement on the sediment surface. Distribution controlled by water column characteristics rather than sediment characteristics.	Pelagic lifestyle/swimming
Smooth exterior, broad & flat axial lobe	Reduce friction	Burrowing
Larger muscle attachments	Stronger limb motions to move sediments	
Eyes reduced or absent	Darker conditions, less need/no need for keen eyesight	Living in deep water
Wide bodies, genal spines		
	Added support for soupy substrate	
Well-developed limbs, flexible hypostome, Cruziana feeding traces	Food passed anteriorly towards the mouth during the course of movement, flexible hypostome used as a scoop.	Particle feeding
Unusual occipital angle, pitted fringe	Pits allow water to flow through cephalon from leg-generated currents	Filter feeding
Rigid, strongly braced hypostome; forked hypostome projections	Ability to process relatively large food particles	Predatory, feeding on soft bodied worms

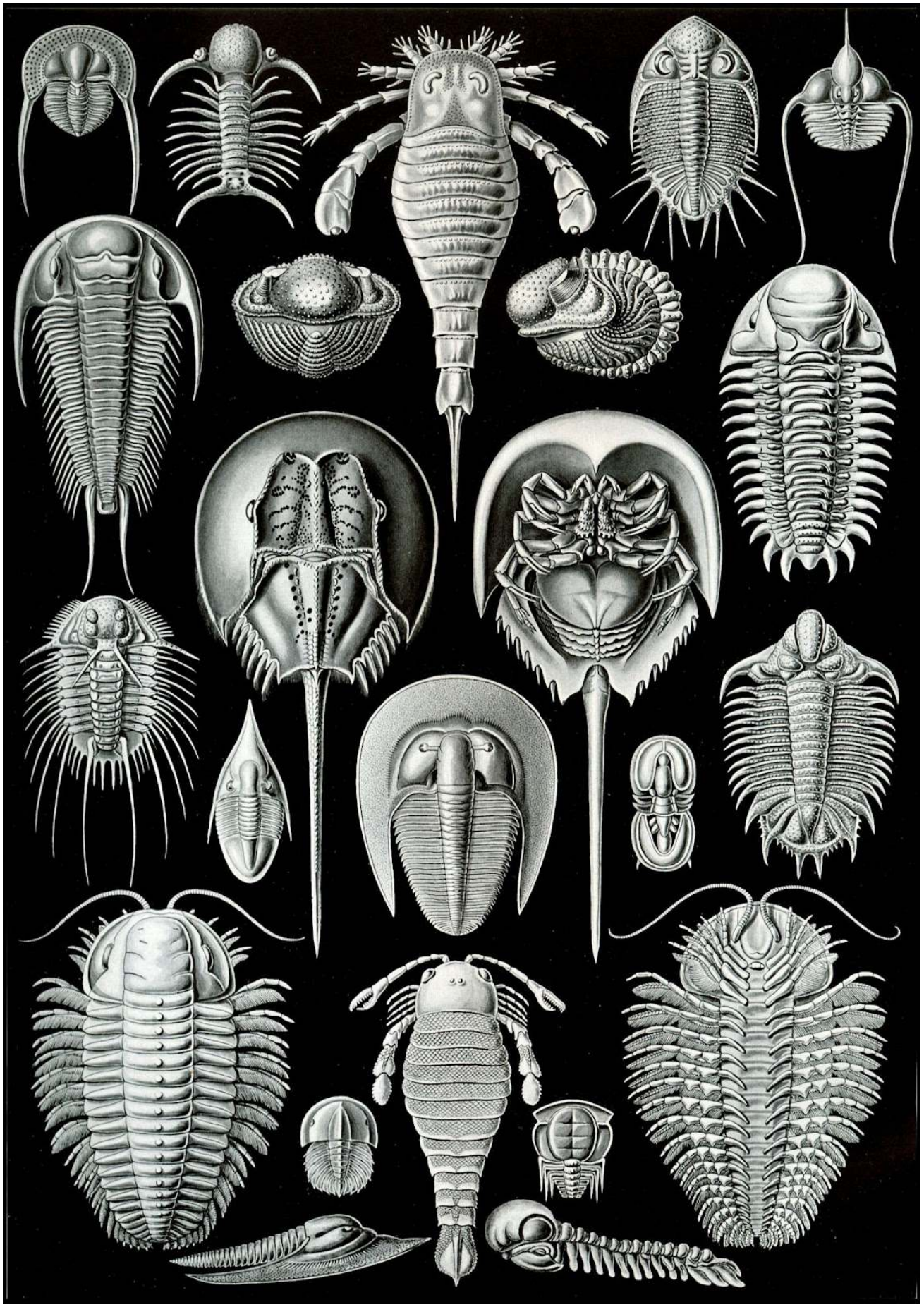


Trilobites task 4: In front of you is the iconic trilobite *Trinucleus*. Given that this dude(-tte) is pretty tiny - and disarticulated - I've put an image of a complete representative to the left to help you out. Using the mode of life table, what do you think the mode of life of this trilobite was?

Trilobites task 5: Find the arthropods on the tree of life in the lecture slides. What is the sister group to Arthropoda? Also name four clades to which the arthropods belong.

Trilobites came in all shapes and sizes, as you can see in the image on the next page, originally published in Ernst Haeckel's *Kunstformen der Natur*. We might say, using the technical language, that they have a high *morphological disparity.* Take a look and get a feel for the range of shapes and sizes you see in the group. But be aware, everything is not as it seems...

Trilobites task 6: Haeckel only had so much space in his book, and so he snuck in some imposters. There are four images of non-trilobites in this figure. I'd like you to spot the odd ones out here, and why not have a punt at figuring out what they are? You can mark the odd ones out by shading them - or however else you want.



Echinoderms

We briefly met the major echinoderm groups in the lecture - in order to keep to time, I had to cut out some of the more interesting extinct groups. Sorry about that! The link on the right will take you to the web page I created with 3D models of the major groups, if you want to quickly revise this, as well as introducing one of the extinct groups (and if you want the full lowdown, the videos on that page talk about the origins of the group and evolution of pentamery [Video #1], as well as some of the major extinct groups [video #2]).

In front of you, you have the Devonian echinoderm *Cupressocrinus* or the Silurian species *Sagenocrinites*. Check it out!

Echinoderm task 1: Of which echinoderm group is the species in front of you a member? Why? See which elements of the anatomy you can identify.

Link *Echinoderms*

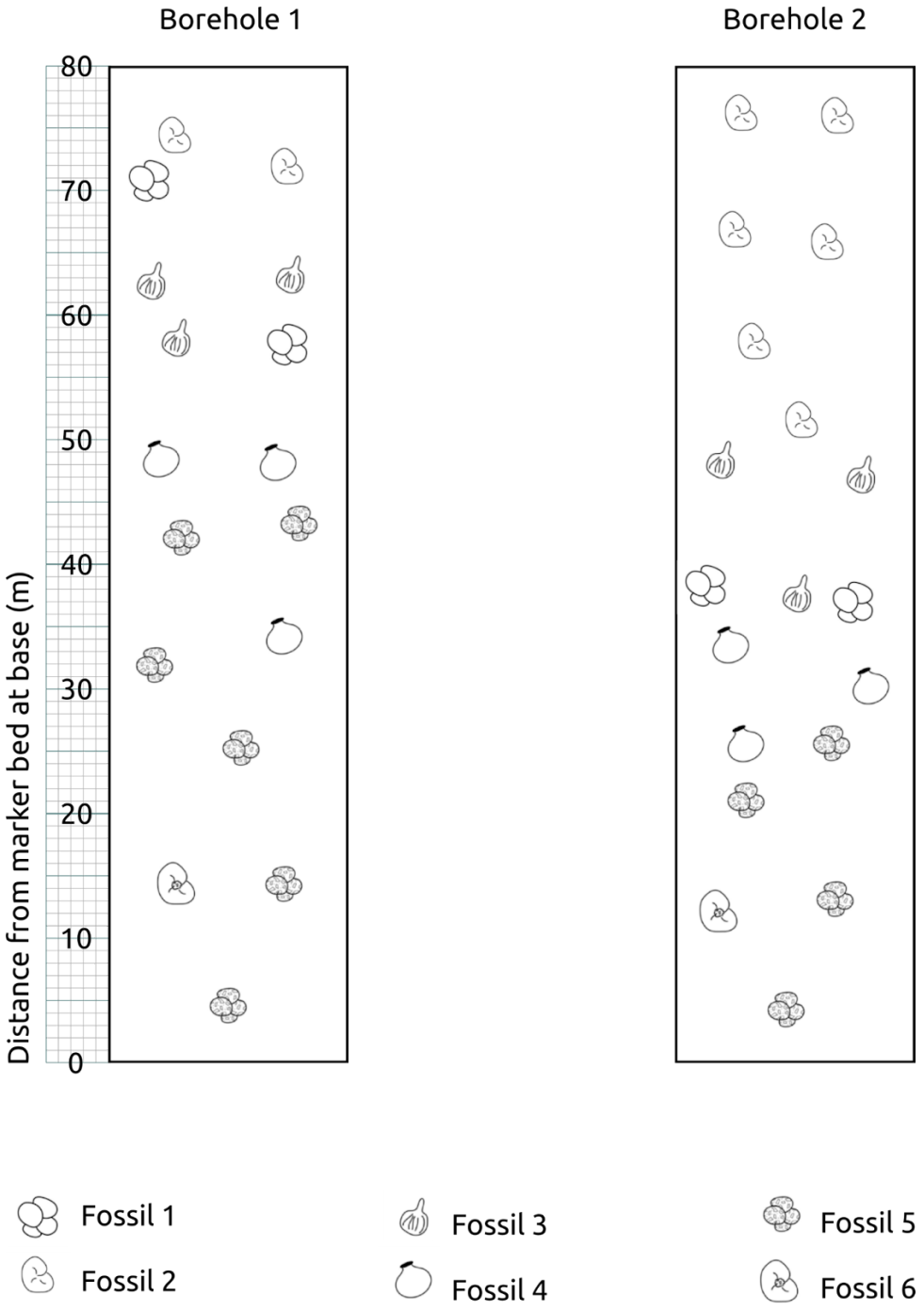
We also learned briefly in the lecture that some groups of echinoids have convergently evolved to be bilaterally symmetrical: this bilateral symmetry overprints their pentameral (i.e. 5-way) symmetry, and creates, for example, a heart-shaped individual. This is associated with a shift in the mode of life of these animals from living on the sediment surface to burrowing (i.e. they move from epi- to infaunal), and as such, there are a whole suite of changes to the morphology in these groups associated with this shift in their mode of life.

Echinoderm task 2: You have access to an example of a regular, and an irregular echinoid. Cool, huh? Study these carefully and see how many changes you can spot associated with the shift from regular to irregular echinoid. *Top tip: There are some changes you won't be able to see with these specimens - when you're done you can chat with a GTA or check out the figure on the lecture slide regarding irregular echinoids that sums up the changes.*

Echinoderm task 3: Members of which phylum are the closest living relatives to the Echinodermata? (If you're interested, this phylum and the Echinodermata are in a clade called the Ambulacraria; Ambulacraria is the sister group to the chordates, which includes all vertebrates).

Biostratigraphy 2

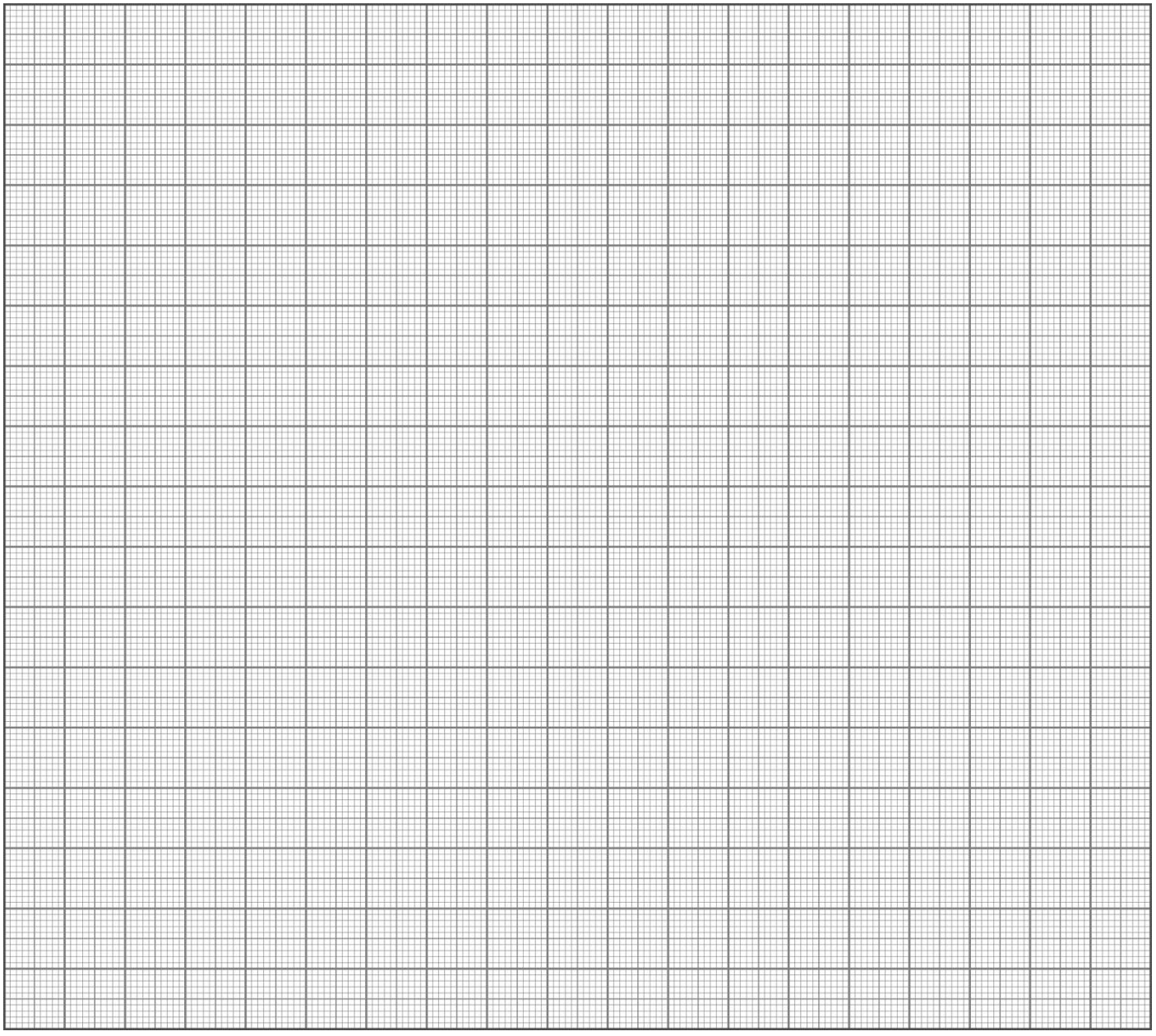
We're moving into a brave new world where there are no useful lithostratigraphic boundaries in our boreholes to help us. I'm afraid we're going to need to use correlation. The two columns below show some microfossil assemblages from two boreholes, in a totally different part of the world. Marked on the left hand side is a scale in metres, in terms of their stratigraphic distance from a marker bed that the two boreholes have in common (and represents a common point in time).



Graphical correlation task 1: Identify the FADS and LADS of each fossil species in m from the marker bed, and add them to the table on the next page. *Note: Whether you measure the "appearance" of the fossil from the bottom, middle or top of each on the image doesn't matter - just make sure you are consistent.*

Borehole 1	FAD	LAD		Borehole 2	FAD	LAD
Fossil 1				**Fossil 1**		
Fossil 2		-NA-		**Fossil 2**		-NA-
Fossil 3				**Fossil 3**		
Fossil 4				**Fossil 4**		
Fossil 5				**Fossil 5**		
Fossil 6				**Fossil 6**		

Graphical correlation task 2: Use the graph paper below to create a graphical correlation between these two deposits - all you have to do is plot the FADS of Borehole 1 (X-axis) against the FADS of Borehole 2 (Y-axis). You can then do the same for the LADS, and draw a best fit line. Note that each pair of FADS and LADS just represent equivalent points in time, so you don't need a separate best fit line for each - one for both will do nicely. The extra page available as a handout also explains how to create this graph. Note LAD for fossil 2 has little meaning as it is found at the top of the borehole.

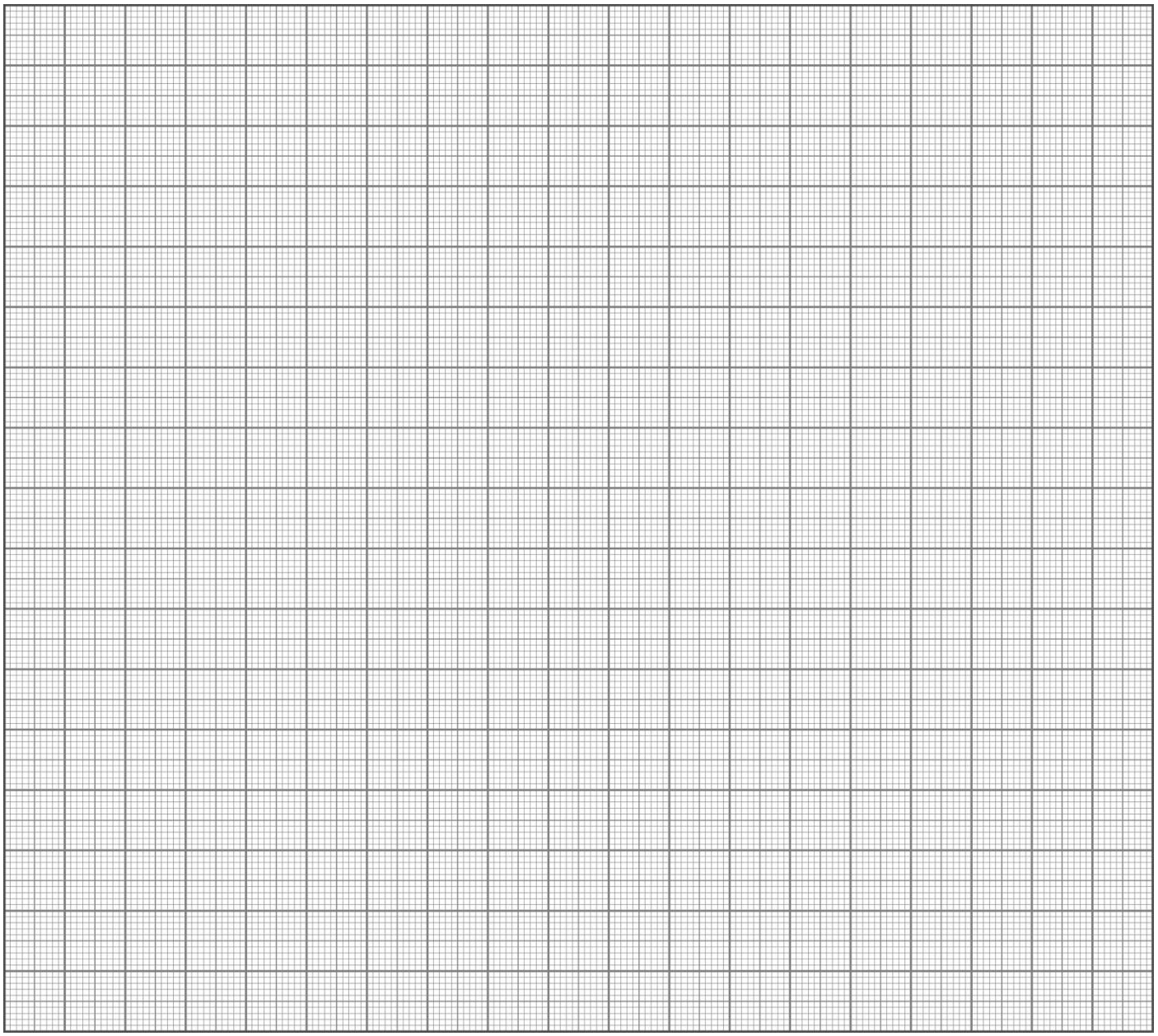


What does this tell us about the relative rates of sedimentation in boreholes one and two?

The table below shows the FADS and LADS of fossils from two boreholes, again shown in metres above a distinctive marker bed at the base of the sedimentary sequence:

Borehole 1	FAD	LAD		Borehole 2	FAD	LAD
Fossil A	0	16		**Fossil A**	0	15
Fossil B	7	28		**Fossil B**	8	28
Fossil C	18	47		**Fossil C**	17	40
Fossil D	37	50		**Fossil D**	38	40
Fossil E	45	65		**Fossil E**	40	63
Fossil F	40	68		**Fossil F**	40	67
Fossil G	60	69		**Fossil G**	48	70

Graphical correlation task 3: By creating a graphical correlation between these two deposits work out the relative sedimentation history between borehole 1 and 2. Once you have marked the points on the graph you can join them up with a line. This doesn't have to consist of one straight segment, but when there are clear changes in sedimentation between the two boreholes you may choose to split it into sensible straight segments at different angles.



What does this graph tell us about the relative sedimentation history between boreholes one and two?

Sources

Allaby, M., 2013. A dictionary of geology and earth sciences. Oxford University Press. (Definitions)

Benton, M.J. and Harper, D.A., 2010 & 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons. (Diagrams, unless otherwise noted)

Casey, M.M. and Lieberman, B.S., 2014. Beyond memorization: an intermediate-level paleontology activity that integrates anatomy, ecology, and macroevolutionary theory using trilobites. Evolution: Education and Outreach, 7(1), pp.1-10. (Trilobite modes of life)

Haeckel, E., 1904. Kunstformen der Natur. Die einhundert Tafeln. (Trilobite plate)